

Quiz 1 study bank

Chapter 1 + Lab 1 — Friday, August 28

The Friday quiz will have a **few** of these questions. Work the set. This first quiz will be graded on **participation**, not correctness.

Reading: *Hundred-Page Language Models Book* Ch. 1.

Lab: [Lab 1](#) notebook.

From the Reading:

1. **[Concept]** Consider the general model $y = f(x)$. What are x , y , and f in supervised learning? (§1.2)
2. **[Concept]** List the four steps of the supervised ML process. (§1.3)
3. **[Concept]** Why do we need a loss function? Give one example of a loss function that is appropriate for linear regression. (§1.2–§1.3)
4. **[Math]** Rewrite $y = wx + b$ for two features using vector notation. (§1.4)
5. **[Math]** For $[1, 0]$ and $[0, 1]$: compute cosine similarity and say whether the vectors point the same way. (§1.4)
6. **[Concept]** Are “parameters” and “weights” used interchangeably in this chapter? What are the parameters in $y = wx + b$? (§1.2)
7. **[Concept]** In logistic regression, consider the standard meaning of z , $\hat{p}$, $\hat{y}$, and y . Which is the logit, the predicted probability, the predicted label, and the observed label? (§1.7–§1.8 + Lab 1)
8. **[Math]** Write the logistic model in standard notation and the sigmoid $\sigma(z)$. (§1.7)
9. **[Math]** Write the equation for binary cross-entropy for one example. (§1.7)
10. **[Concept]** Why do we have to use iterative optimization (like gradient descent) for logistic regression? (§1.7)

Lab / code

11. [Code] What does `loss.backward()` compute, and what does `optimizer.step()` do next?

12. [Code] In

```
model = nn.Sequential(  
    nn.Linear(2, 1),  
    nn.Sigmoid()  
)
```

what do 2 and 1 represent? Which tensor is z , and which is $\hat{p}$? How do you get $\hat{y}$ from $\hat{p}$?

13. [Math] Suppose we have `layer = nn.Linear(12, 1)`. How many **trainable** numbers (weights + bias) are in this layer? If the input has shape `(32, 12)`, what is the output shape?

14. [Lab] Suppose that network has almost zero training loss on a tiny dataset. Name a potential problem.